

Re: Application for the EngD position, Developing an Interactive Reverse Logistics Map for Circular Construction (CIRCOLOGIC), University of Twente

Dear Professor Hans Voordijk, Professor Berkhout, Professor Marc, and Members of the Selection Committee,

Circular construction's hardest problem is not physically recovering materials; it is deciding, quickly and credibly, whether reuse makes sense once availability, quality, timing, transport, storage, cost, and liability are all uncertain. That is fundamentally an information and decision problem, which is precisely why CIRCOLOGIC's design resonates with me: an interactive map that makes donor and target materials visible, a calculation module that quantifies the CO2, cost, and circularity trade-offs, and a scenario engine that lets actors compare deconstruction, reuse, refurbishment, modular, and new-build options. Building that triad is the kind of work I want to commit the next stage of my career to.

My Master's thesis on Agile project management in construction convinced me that the sector's transformation depends as much on coordination and transparent information flow as on any single technology. Following that thread to the end of a building's life raised a question I have not been able to put down: reusable material is lost less because we cannot recover it than because reliable information about it never reaches the people making design and logistics decisions. That shifted my ambition from delivering projects to designing the artifacts that change decisions and evaluating them in practice, which is exactly the design-science character of this EngD.

My experience maps onto each part of the project. Working on digital twins at Aug.e, I integrated IoT, metadata, and BIM data from physical assets into decision-support, the same data-model challenge of aligning planning data, material inventories, product passports, and logistics parameters. At Hive CPQ I encoded complex, constraint-heavy product rules into configurators that non-specialists trusted and used, close in spirit to a calculation and scenario engine that must make trade-offs legible. Consulting on Agile delivery taught me to coordinate the mixed stakeholder networks reuse depends on, from contractors and designers to logistics providers, recyclers, and municipalities, which is what requirements definition with Dura Vermeer, UT, and TNO will demand. With a civil and environmental engineering foundation and GIS skills for the spatial layer, I can contribute to both the build and its adoption.

Most recently I have taken a deliberate break to care for my newborns, keeping my thinking current through self-study in circular construction, reverse logistics, and GIS. I see this EngD as the ideal way back: a setting where I can not only recover the ground I built earlier, but deepen it and, crucially, test it against a real industry problem rather than in theory.

The University of Twente's engineering-for-impact philosophy, and an EngD that deliberately pairs advanced coursework with an industry design project embedded in ECOLOGIC, PACER, and the UT and TNO Joint Innovation Center, is the environment I have been looking for. I do my best work when new methods are put to use immediately and refined with partners, and I am comfortable moving between research groups, contractors, and institutes to get there.

I would be honoured to help build CIRCOLOGIC and to grow into an engineer who turns circular ambitions into tools that people actually use. I would welcome the chance to discuss how my background and motivation fit the project, and I thank you for your consideration.

Yours sincerely,
Pooja Awasare